

External Tools and Resources Statement

MACS 40550: Agent-Based Modeling Spatial Structure and the Mobilization Consequences of Repression: An Agent-Based Model of Protest Dispersal

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<https://github.com/asttanns/abm-spatial-repression>

Mesa Official Documentation (<https://mesa.readthedocs.io/stable/>)

- Primary reference for Mesa 3.5.1 overall architecture, including Agent and Model base classes, MultiGrid construction, and DataCollector setup
- Consulted for `agents.shuffle_do()` activation method and Mesa 3.x migration guide, particularly the deprecation of scheduler-based activation
- Consulted for AgentPortrayalStyle API in `mesa.visualization.components`, replacing the older dict-return portrayal form

Mesa Space API Reference (<https://mesa.readthedocs.io/stable/apis/space.html>)

- Consulted for MultiGrid construction, including `torus=False` for hard grid boundaries, and how `get_neighborhood`, `get_cell_list_contents`, `move_agent`, and `coord_iter` operate within a MultiGrid
- Clarified how `move_agent` updates agent position within a MultiGrid to translate dispersal mechanism
- Consulted for `coord_iter` usage in `_check_cell_capacity`, which iterates over all cells and their contents

Mesa BatchRunner Documentation (<https://mesa.readthedocs.io/stable/apis/batchrunner.html>)

- Consulted for `batch_run` parameter structure and output DataFrame format
- Used to diagnose seed-handling behavior in Mesa 3.5.1

Claude (Anthropic)

- Used throughout code development as a debugging partner, especially for diagnosing the `coord_iter` unpacking error in `_check_cell_capacity()`, the deprecated `seed` vs `rng` parameter issue, and the AgentPortrayalStyle import failure in Mesa 3.5.1
- Consulted for Mesa 3.x API compatibility questions during model iteration
- Used to assist the batch run implementation
- Used to assist the iteration for analysis script used to visualize batch run result (`analysis.py`)